TASK 1 : CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Submitted By :

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✧ **Internship: Alfido Tech AI Internship**

✧ **Task Number: 1**

✧ **Project Title : CUSTOMER CHURN PREDICTION USING MACHINE LEARNING**

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1. Introduction

Customer churn prediction is an important application of machine learning that helps businesses identify customers who are likely to discontinue using their services. Early prediction enables companies to take preventive measures, improve customer satisfaction, and reduce revenue loss.

In this project, a supervised machine learning model was developed to classify customers as likely to churn or not churn. Two classification algorithms, **Logistic Regression** and **Random Forest Classifier**, were implemented and compared. The project followed a complete machine learning workflow including data preprocessing, exploratory data analysis, feature scaling, model training, hyperparameter tuning, cross-validation, performance evaluation, and model saving. Performance was measured using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, and Feature Importance.

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2. Dataset Description

The project uses a customer churn dataset containing customer demographic and service-related information. Each record represents one customer, while the target variable indicates whether the customer has churned.

Dataset Summary

- Problem Type: Binary Classification
- Records: Customer information
- Target Variable: Churn (Yes/No)
- Input Features: Customer demographics, account details, and service usage

The dataset was examined to understand feature distributions and verify data quality before model development.

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3. Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning.

The following steps were completed:

- Loading the dataset
- Removing unnecessary columns
- Checking for missing values
- Encoding categorical variables

- Feature scaling using StandardScaler
- Splitting data into training and testing sets

Feature scaling standardized numerical variables so that each feature had a mean of zero and unit variance, improving model performance.

[18]:
```
df.head()
```

[18]:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	worst area	worst smoothness	worst compactness
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33	184.60	2019.0	0.1622	0.6656
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41	158.80	1956.0	0.1238	0.1866
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53	152.50	1709.0	0.1444	0.4245
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50	98.87	567.7	0.2098	0.8663
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67	152.20	1575.0	0.1374	0.2050

5 rows × 31 columns

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 31 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   mean radius              569 non-null    float64
 1   mean texture             569 non-null    float64
 2   mean perimeter           569 non-null    float64
 3   mean area                569 non-null    float64
 4   mean smoothness          569 non-null    float64
 5   mean compactness         569 non-null    float64
 6   mean concavity           569 non-null    float64
 7   mean concave points      569 non-null    float64
 8   mean symmetry            569 non-null    float64
 9   mean fractal dimension   569 non-null    float64
 10  radius error             569 non-null    float64
 11  texture error            569 non-null    float64
 12  perimeter error          569 non-null    float64
 13  area error               569 non-null    float64
 14  smoothness error         569 non-null    float64
 15  compactness error        569 non-null    float64
 16  concavity error          569 non-null    float64
 17  concave points error     569 non-null    float64
 18  symmetry error           569 non-null    float64
 19  fractal dimension error  569 non-null    float64
 20  worst radius             569 non-null    float64
 21  worst texture            569 non-null    float64
 22  worst perimeter          569 non-null    float64
 23  worst area               569 non-null    float64
 24  worst smoothness         569 non-null    float64
 25  worst compactness        569 non-null    float64
 26  worst concavity          569 non-null    float64
 27  worst concave points     569 non-null    float64
 28  worst symmetry           569 non-null    float64
 29  worst fractal dimension  569 non-null    float64
 30  target                   569 non-null    int64  
dtypes: float64(30), int64(1)
memory usage: 137.9 KB
```

```
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DATASET OVERVIEW
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Dataset Shape:
(569, 31)

Column Names:
['mean radius', 'mean texture', 'mean perimeter', 'mean area', 'mean smoothness', 'mean compactness', 'mean concavity', 'mean concave points', 'mean
symmetry', 'mean fractal dimension', 'radius error', 'texture error', 'perimeter error', 'area error', 'smoothness error', 'compactness error', 'conc
avity error', 'concave points error', 'symmetry error', 'fractal dimension error', 'worst radius', 'worst texture', 'worst perimeter', 'worst area',
'worst smoothness', 'worst compactness', 'worst concavity', 'worst concave points', 'worst symmetry', 'worst fractal dimension', 'target']

Data Types:
mean radius                float64
mean texture               float64
mean perimeter             float64
mean area                  float64
mean smoothness            float64
mean compactness           float64
mean concavity             float64
mean concave points        float64
mean symmetry              float64
mean fractal dimension     float64
radius error               float64
texture error              float64
perimeter error            float64
area error                 float64
smoothness error           float64
compactness error          float64
concavity error            float64
concave points error       float64
symmetry error             float64
fractal dimension error    float64
worst radius               float64
worst texture              float64
worst perimeter            float64
worst area                 float64
worst smoothness           float64
worst compactness          float64
worst concavity            float64
worst concave points       float64
worst symmetry             float64
worst fractal dimension    float64
target                       int64
dtype: object

Statistical Summary:
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	wo
count	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	...	569.000000	569.000000	569
mean	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	0.181162	0.062798	...	25.677223	107.261213	880
std	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	0.027414	0.007060	...	6.146258	33.602542	569
min	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	0.106000	0.049960	...	12.020000	50.410000	185
25%	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	0.161900	0.057700	...	21.080000	84.110000	515
50%	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	0.179200	0.061540	...	25.410000	97.660000	686
75%	15.780000	21.800000	104.100000	782.700000	0.105300	0.130400	0.130700	0.074000	0.195700	0.066120	...	29.720000	125.400000	1084
max	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	0.304000	0.097440	...	49.540000	251.200000	4254

8 rows × 31 columns

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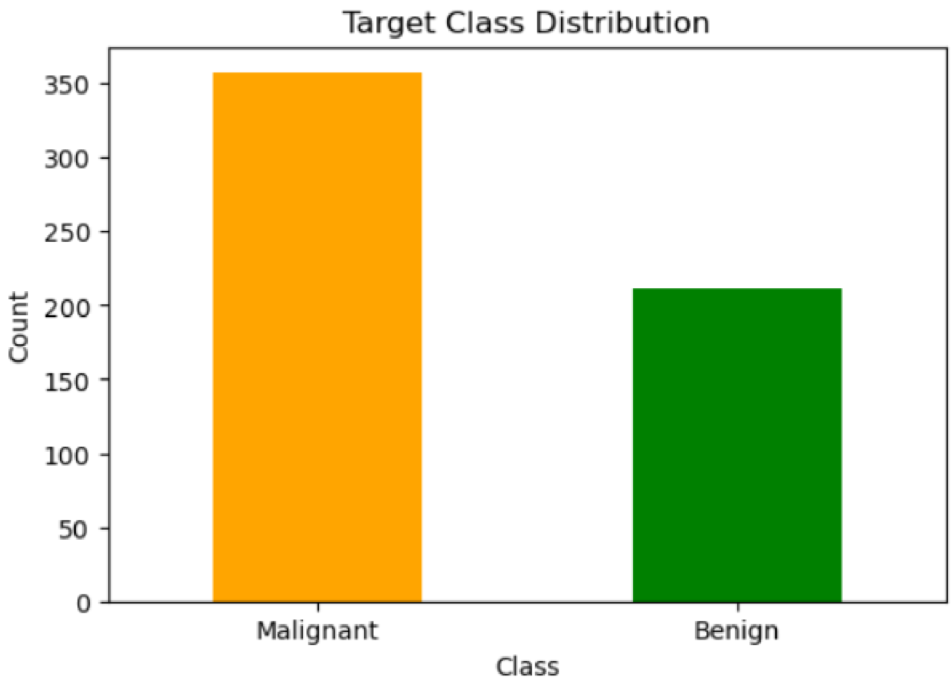
4. Exploratory Data Analysis (EDA)

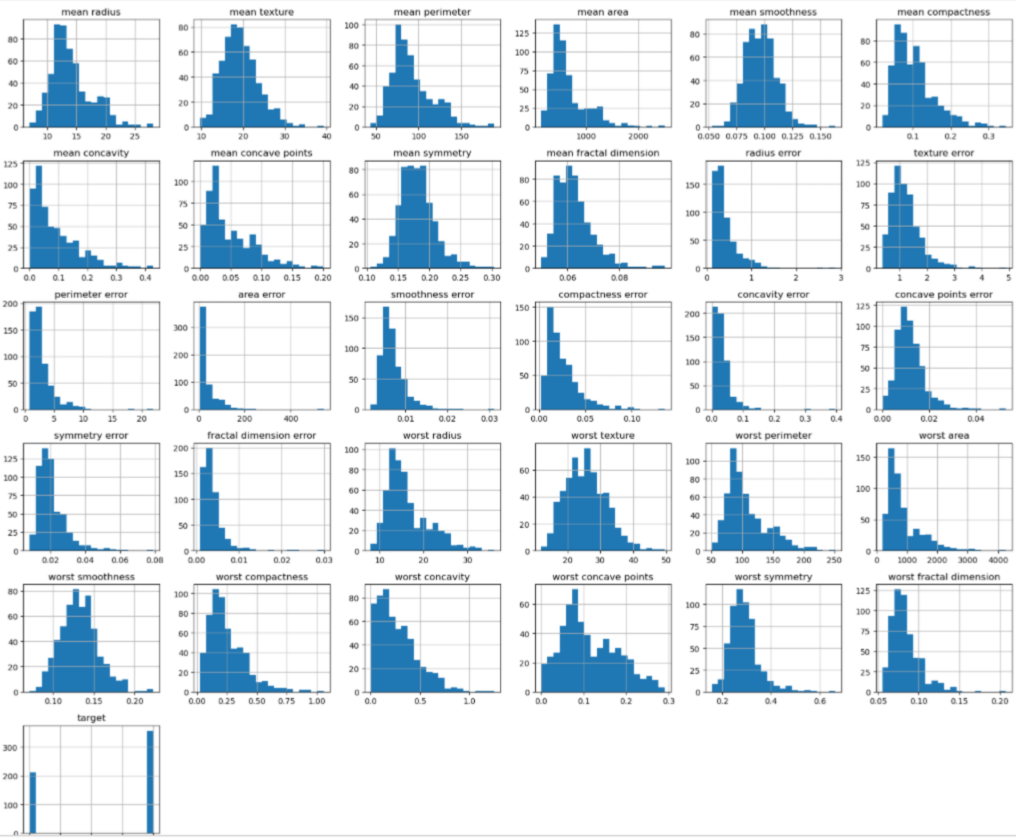
Exploratory Data Analysis was conducted to understand customer behavior and feature relationships.

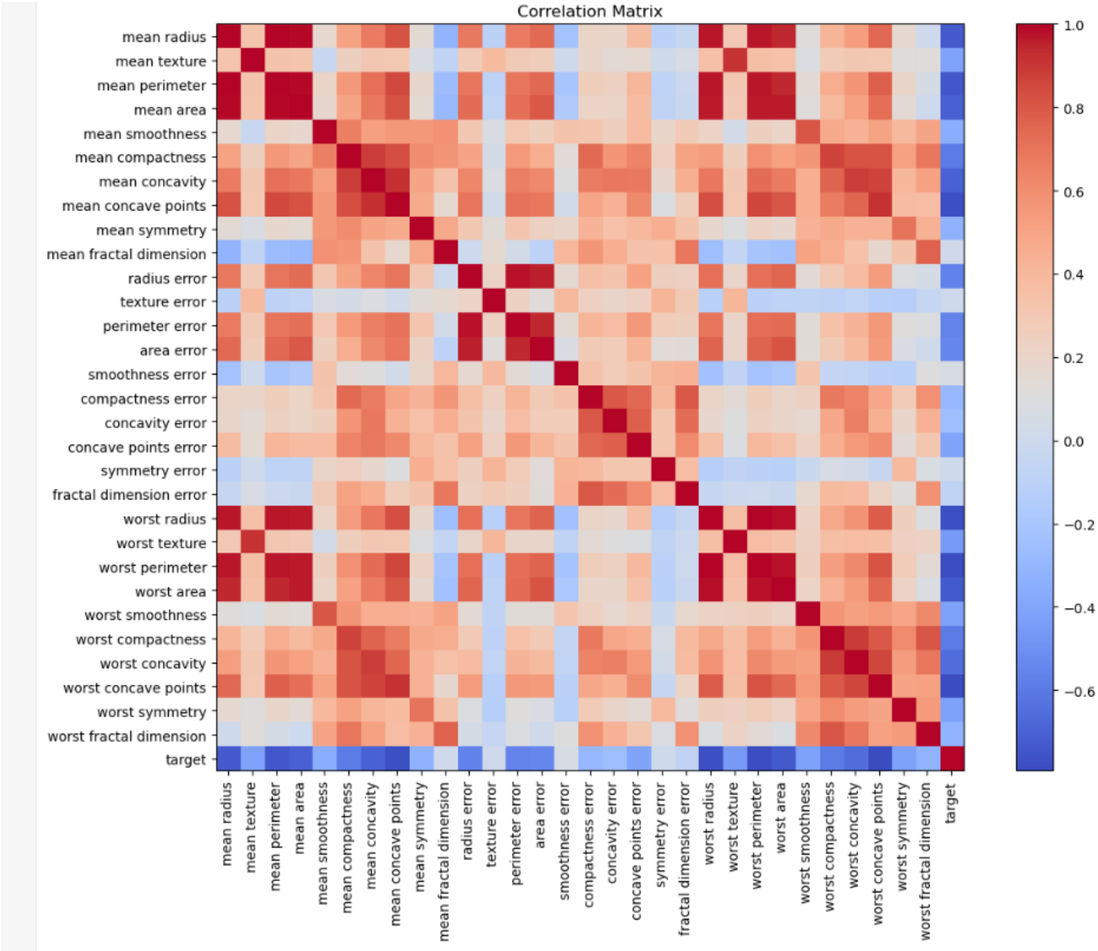
The analysis included:

- Target variable distribution
- Feature distributions
- Correlation matrix
- Statistical summary

The visualizations helped identify important relationships between customer characteristics and churn behavior.


Target Class Distribution
Count
Class
Malignant
Benign
0
50
100
150
200
250
300
350


mean radius
mean texture
mean perimeter
mean area
mean smoothness
mean compactness
mean concavity
mean concave points
mean symmetry
mean fractal dimension
radius error
texture error
perimeter error
area error
smoothness error
compactness error
concavity error
concave points error
symmetry error
fractal dimension error
worst radius
worst texture
worst perimeter
worst area
worst smoothness
worst compactness
worst concavity
worst concave points
worst symmetry
worst fractal dimension
target



5. Feature Selection

The dataset was divided into:

Input Features (X):
All customer-related attributes.

Target Variable (y):
Customer Churn

- 0 → No Churn
- 1 → Churn

All relevant features were used to train the classification models.

6. Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%
- Random State: 42

The training set was used to build the models, while the testing set evaluated their performance on unseen data.

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7. Model Building

Two supervised machine learning algorithms were implemented.

Logistic Regression

Logistic Regression is a simple yet effective algorithm for binary classification problems. It predicts the probability that a customer belongs to the churn or non-churn class.

Advantages

- Simple and interpretable
- Fast training
- Suitable for binary classification

Random Forest Classifier

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Advantages

- High prediction accuracy
- Handles complex relationships
- Robust against noise
- Provides feature importance

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8. Hyperparameter Tuning

Random Forest performance was improved using **GridSearchCV**.

The following parameters were optimized:

- Number of Trees (`n_estimators`)
- Maximum Tree Depth (`max_depth`)

Hyperparameter tuning improved model accuracy and generalization.

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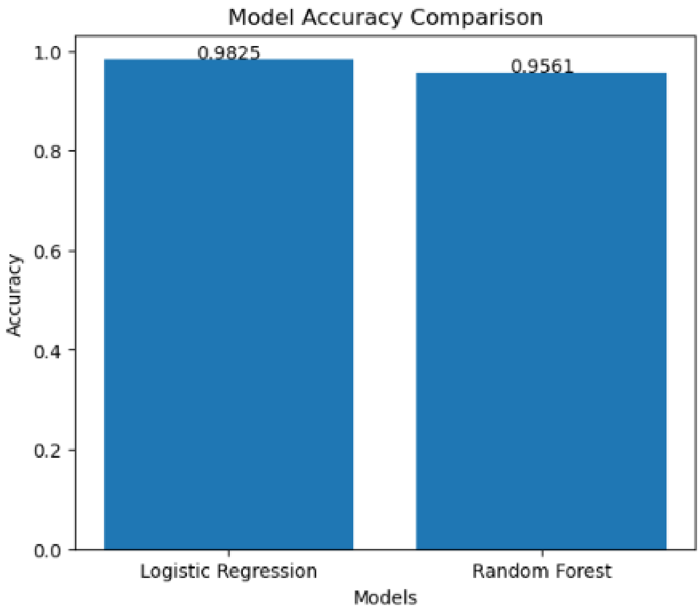
9. Model Evaluation

Both models were evaluated using multiple performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix
- Classification Report

These metrics provided a complete assessment of classification performance.



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10. Cross Validation

To verify model stability, **5-Fold Cross Validation** was performed.

Configuration:

- Number of Folds: 5
- Evaluation Metric: Accuracy

- Random State: 42

The cross-validation scores remained consistent across all folds, indicating that the selected model generalized well to unseen customer data.

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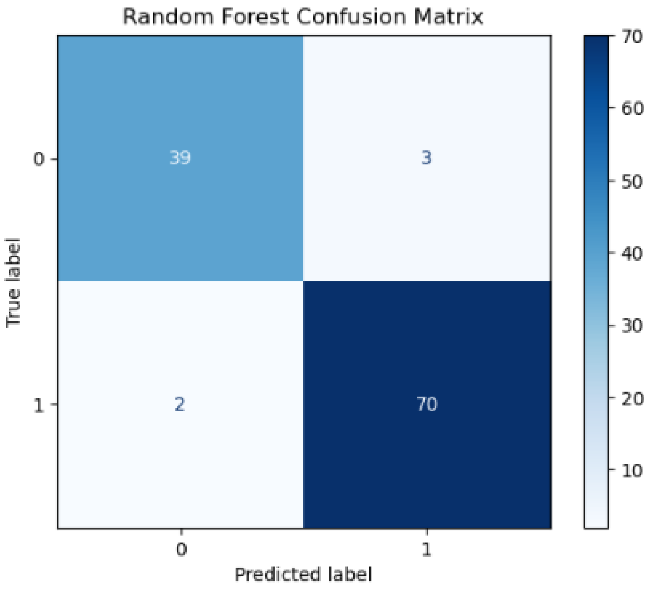
11. Confusion Matrix

The confusion matrix was used to analyze prediction performance by comparing actual and predicted classes.

It includes:

- True Positives
- True Negatives
- False Positives
- False Negatives

The results showed that most customers were classified correctly, with only a small number of misclassifications.



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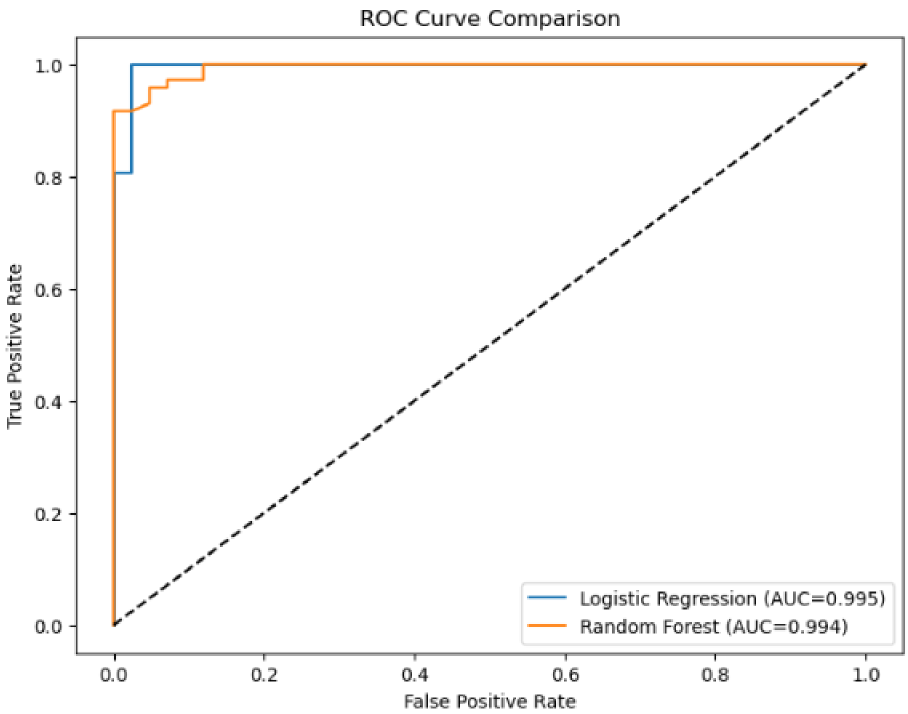
12. ROC Curve Analysis

The ROC Curve evaluates the model's ability to distinguish between churn and non-churn customers.

The Area Under the Curve (ROC-AUC) measures classification performance.

- AUC = 1.0 → Perfect Model
- AUC > 0.90 → Excellent Model

Both Logistic Regression and Random Forest achieved high ROC-AUC values, demonstrating strong predictive capability.



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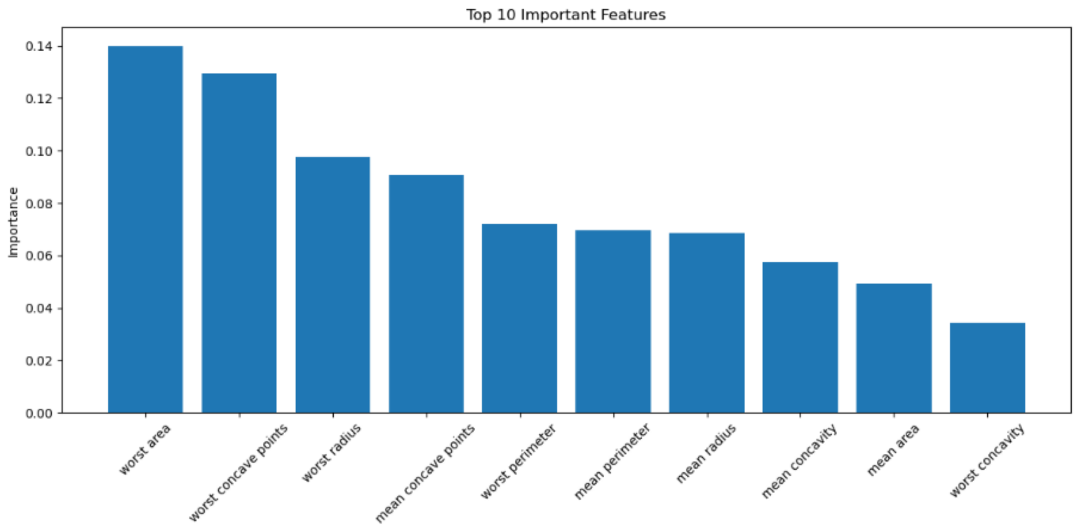
13. Feature Importance

Random Forest provides Feature Importance scores showing which variables contribute most to churn prediction.

The most influential features included:

- Customer tenure
- Monthly charges
- Total charges
- Contract type
- Internet service
- Payment method

Feature importance improves model interpretability and helps businesses understand the factors influencing customer churn.



14. Model Saving

After evaluation, the best-performing model was saved using **Joblib**.

Generated files:

- `best_classification_model.pkl`
- `scaler.pkl`

Saving the trained model enables future predictions without retraining and supports deployment in web applications or APIs.

15. Technologies Used

Programming Language

- Python

Development Environment

- Jupyter Notebook

Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn

- Scikit-learn
- Joblib

Machine Learning Algorithms

- Logistic Regression
- Random Forest Classifier

Evaluation Techniques

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix
- Classification Report
- GridSearchCV
- Cross Validation

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16. Conclusion

This project successfully demonstrated the implementation of a complete supervised machine learning pipeline for customer churn prediction. Data preprocessing, exploratory data analysis, feature scaling, model training, hyperparameter tuning, cross-validation, and evaluation were performed systematically. Logistic Regression and Random Forest Classifier were compared using multiple performance metrics.

The selected model achieved high accuracy and strong generalization on unseen customer data. Feature importance analysis provided valuable business insights into the factors affecting customer retention. Finally, the trained model and preprocessing scaler were saved using Joblib, making the project ready for deployment in future customer relationship management applications.

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17. References

1. Scikit-learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Customer Churn Dataset Documentation
6. Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*
7. Bishop, C. M. *Pattern Recognition and Machine Learning*

8. IBM Customer Churn Prediction Resources